

Computer Vision SS26 - Exercise 7

Jun. Prof. Dr. Kira Maag

March 2026

This exercise will be discussed the week of 29th of June.

1 Introduction

The following exercises cover global stereo matching, disparity space images, Markov Random Fields, and smoothness priors for depth estimation.

2 Exercises

Exercise 1: Disparity Space Image

Consider a rectified stereo pair. For one horizontal scanline, the left and right images have width 6 pixels. A possible match between left pixel i and right pixel j has disparity

$$d = i - j.$$

Assume valid disparities satisfy

$$0 \leq d \leq 2.$$

[label=2.]

1. Draw a 6×6 disparity space image (DSI), where the rows correspond to right-image positions j and the columns to left-image positions i .
2. Mark all invalid cells caused by the constraints $d < 0$ and $d > 2$.
3. For the valid matches

$$(i, j) \in \{(1, 1), (2, 1), (3, 2), (4, 3), (5, 3), (6, 4)\},$$

compute the disparities.

4. Explain why finding a good stereo matching for a whole scanline can be formulated as finding a low-cost path through the DSI.

Exercise 2: Dynamic Programming for Stereo Matching

We want to find the minimum-cost path through the following cost matrix:

$$C = \begin{pmatrix} 1 & 4 & 6 & 8 \\ 3 & 2 & 5 & 7 \\ 6 & 3 & 2 & 4 \\ 9 & 6 & 3 & 1 \end{pmatrix}.$$

A path starts at the upper-left cell and ends at the lower-right cell. From each cell, the allowed moves are right, down, or diagonal down-right.

[label=2.]

1. Write down the dynamic programming recurrence for the accumulated minimum cost $D(i, j)$.
2. Compute the complete table of accumulated costs D .
3. What is the minimum path cost?
4. Recover one minimum-cost path and briefly interpret diagonal, horizontal, and vertical moves in the context of stereo matching.

Exercise 3: Markov Random Field for Disparity Smoothing

Let a scanline contain four pixels with unknown disparities

$$D = (d_1, d_2, d_3, d_4),$$

where each disparity can take values in $\{0, 1, 2\}$.

The energy of a disparity assignment is

$$E(D) = \sum_{i=1}^4 \psi_{\text{data}}(i, d_i) + \lambda \sum_{i=1}^3 \psi_{\text{smooth}}(d_i, d_{i+1}),$$

with $\lambda = 2$ and

$$\psi_{\text{smooth}}(a, b) = \begin{cases} 0, & a = b, \\ 1, & a \neq b. \end{cases}$$

The unary data costs are

	$d = 0$	$d = 1$	$d = 2$
$i = 1$	0	2	4
$i = 2$	1	0	3
$i = 3$	3	0	1
$i = 4$	4	2	0

[label=2.]

1. Compute the energy of the assignment

$$D = (0, 1, 1, 2).$$

2. Compute the energy of the assignment

$$D = (1, 1, 1, 1).$$

3. Which of the two assignments is preferred by the MRF energy?
4. Explain the role of the parameter λ . What happens if λ is chosen very large?

Exercise 4: From Probability to Energy

A second-order graphical model for disparity estimation is defined as

$$p(D) \propto \exp \left(- \sum_i \psi_{\text{data}}(d_i) - \lambda \sum_{i \sim j} \psi_{\text{smooth}}(d_i, d_j) \right).$$

[label=2.]

1. Explain what the unary term $\psi_{\text{data}}(d_i)$ represents.
2. Explain what the pairwise term $\psi_{\text{smooth}}(d_i, d_j)$ represents.
3. Show that maximizing $p(D)$ is equivalent to minimizing an energy function $E(D)$.
4. Why is this model useful for depth estimation from stereo images?